

Multi-Task Dependency Problem

Problem Statement

A city is planning a transportation network upgrade. The project involves: 1. Analyzing current traffic patterns 2. Designing new routes 3. Estimating construction costs 4. Optimizing budget allocation

Each subtask depends on previous results.

Task Breakdown

Task 1: Traffic Analysis - Collect traffic data from 10 intersections - Identify peak congestion times - Calculate average wait times

Task 2: Route Design - Depends on: Task 1 - Propose 3 alternative routes - Estimate travel time improvements - Consider environmental impact

Task 3: Cost Estimation - Depends on: Task 2 - Calculate construction costs for each route - Include maintenance estimates - Factor in land acquisition costs

Task 4: Budget Optimization - Depends on: Task 3 - Allocate budget across routes - Prioritize based on cost-benefit ratio - Ensure 5-year ROI

Dependencies

Task 1 → Task 2 → Task 3 → Task 4

Objectives

- Complete all tasks in dependency order
- Find optimal budget allocation
- Maximize network efficiency